

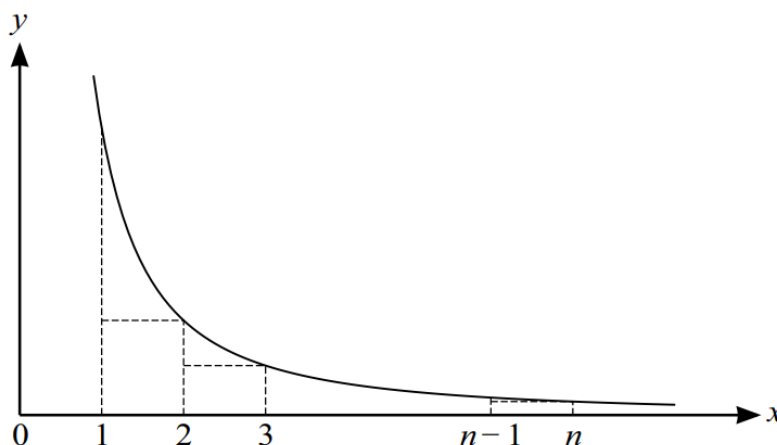
PAST EXAMINATION QUESTIONS (FP2)

TOPIC 12 : INTEGRATION (1) – TECHNIQUES AND RECTANGULAR APPROXIMATION

1. Find the exact value of $\int_0^1 \frac{1}{\sqrt{(3+4x-4x^2)}} dx$. $\left(\frac{\pi}{6}\right)$ [6]

Specimen Paper 2020/2/2

2.



The diagram shows the curve with equation $y = \frac{1}{x^2}$ for $x > 0$, together with a set of $(n - 1)$ rectangles of unit width.

- (a) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{1}{r^2} < \frac{2n-1}{n}. \quad [5]$$

- (b) Use a similar method to find, in terms of n , a lower bound for $\sum_{r=1}^n \frac{1}{r^2}$. $\left(1 - \frac{1}{n+1} \text{ OR } 1 - \frac{1}{n} + \frac{1}{n^2}\right)$ [3]

Specimen Paper 2020/2/4

3. (a) Given that $y = \sqrt{\sinh x}$ for $x \geq 0$, express $\frac{dy}{dx}$ in terms of y only. $\left(\frac{\sqrt{1+y^4}}{2y}\right)$ [3]
 (b) Hence or otherwise find $\int \frac{2t}{\sqrt{(1+t^4)}} dt$. $(\sinh^{-1}(t^2) + c \text{ OR } \ln|\sec \theta + \tan \theta| + c \text{ OR } \ln|t^2 + \sqrt{1+t^4} + c|)$ [3]

CAIE 9795/Specimen Paper 2020/1/4

4. (a) Without using a calculator, show that if $\sinh x = 1$ then $\tanh \frac{1}{2}x = \sqrt{2} - 1$. [5]
 (b) Show that $\frac{d}{dx} \left(2 \tan^{-1} \left(\tanh \frac{1}{2}x \right) \right) = \frac{1}{\cosh x}$. [4]
 (c) Use the substitution $\tan \theta = \sinh^2 x$ to find the exact value of

$$\int_0^{\frac{1}{4}\pi} \frac{\sqrt{\tan \theta} \sec^2 \theta}{1 + \tan \theta} d\theta. \quad \left(2 - \frac{1}{2}\pi\right) \quad [7]$$

[You may use, without proof, the result that $\tan \frac{1}{8}\pi = \sqrt{2} - 1$.]

CAIE 9795/May/June 2019/1/12

5. (i) Given that $x \geq 1$, use the substitution $x = \cosh \theta$ to show that

$$\int \frac{1}{x^2 \sqrt{x^2 - 1}} dx = \frac{\sqrt{x^2 - 1}}{x} + C$$

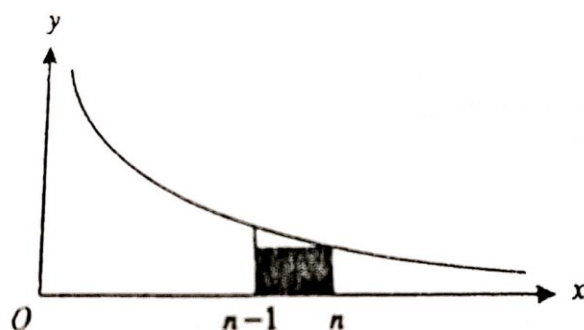
where C is an arbitrary constant. [4]

- (ii) By differentiating $\sec y = x$ implicitly, show that $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2 - 1}}$ for $x \geq 1$. [4]

- (iii) Use integration by parts to determine $\int \frac{\sec^{-1} x}{x^2} dx$ for $x \geq 1$. [4]

CAIE 9795/May/June 2017/1/9

6.



The diagram shows a sketch of the graph of $y = \frac{1}{\sqrt{x}}$. By considering the shaded rectangle and the area of the region between the graph and the x -axis for $n - 1 \leq x \leq n$, show that

$$\frac{1}{\sqrt{n}} < 2\{\sqrt{n} - \sqrt{n-1}\}. \quad [3]$$

Deduce that

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} < 2\sqrt{n}. \quad [2]$$

Show also that

$$\frac{1}{\sqrt{n}} > \{2\sqrt{n+1} - \sqrt{n}\}. \quad [2]$$

Deduce that

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} > 2\sqrt{n+1} - 2. \quad [2]$$

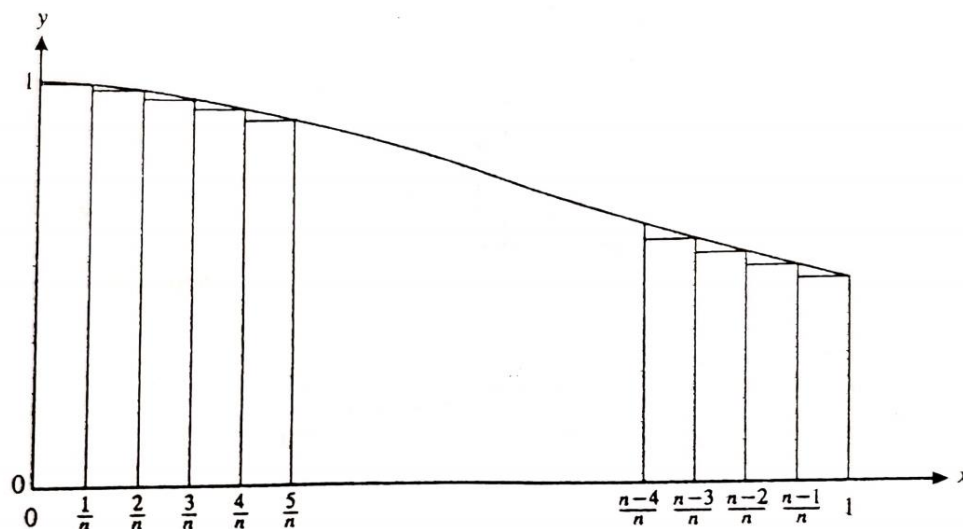
Hence find a value of N for which

[3]

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{N}} > 1000.$$

Syllabus 9233/Nov 1996/1/18

7. (ii)



The graph of $y = \frac{1}{1+x^2}$, for $0 \leq x \leq 1$, is shown in the diagram. Rectangles, each of width $\frac{1}{n}$, are drawn under the curve. Show that the total area A of all n rectangles is given by

$$A = \frac{1}{n} \left\{ \frac{1}{1 + \left(\frac{1}{n}\right)^2} + \frac{1}{1 + \left(\frac{2}{n}\right)^2} + \frac{1}{1 + \left(\frac{3}{n}\right)^2} + \cdots + \frac{1}{2} \right\}. \quad [2]$$

State the limit of A as $n \rightarrow \infty$. [1]

Syllabus 9233/Nov 2001/1/18

8. (b) Integrate $\frac{1}{\sqrt{1-4y^2}}$ with respect to y . $\left(\frac{1}{2}\sin^{-1}(2y) + c\right)$

Nov 1990/1/10

9. Find
(b) $\int \frac{1}{\sqrt{4-9x^2}} dx \cdot \left(\frac{1}{3}\sin^{-1}\left(\frac{3}{2}x\right) + c\right)$

Nov 1989/1/4 (part)

10. (a) Find $\int \frac{1}{\sqrt{2x^2+2x-1}} dx \cdot \left(\frac{1}{\sqrt{2}}\cosh^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + c\right)$ [4]

December 1985/2/16 (part)

11. (i) Prove that $\int_0^{\ln 2} \cosh^2 x \, dx = \frac{15}{32} + \frac{1}{2}\ln 2$. [5]

(ii) Obtain a corresponding result for $\int_0^{\ln 2} \sinh^2 x \, dx$. $\left(\frac{15}{32} + \frac{1}{2}\ln 2\right)$ [3]

(iii) Prove that $\int_0^{\ln 2} \sinh^2 x \cosh^2 x \, dx = \frac{255}{1024} - \frac{1}{8} \ln 2$. [5]

November 1985/1/20

12. Given that $y = \cosh^{-1} x$ and that $y > 0$, show that $y = \ln(x + \sqrt{x^2 - 1})$ and that

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 - 1}}. \quad [4]$$

Verify that, when $t = x + x^{-1}$, then $x^4 + x^2 + 1 = x^2(t^2 - 1)$. Deduce, by using this substitution, or show otherwise, that

$$\int_1^2 \frac{x^2 - 1}{x\sqrt{x^4 + x^2 + 1}} \, dx = \ln \left(\frac{5 + \sqrt{21}}{2 + 2\sqrt{3}} \right). \quad [9]$$

November 1985/2/20

13. (a) Find $\int \sqrt{4 + x^2} \, dx$. $\left(\frac{1}{2} x \sqrt{4 + x^2} + 2 \sinh^{-1} \left(\frac{1}{2} x \right) \right)$ [5]

June 1985/2/17 (part)

14. Prove that $\int_0^1 \sqrt{x^2 + 1} \, dx = \frac{1}{2} [\sqrt{2} + \ln(1 + \sqrt{2})]$.

MB June 1985/1/20 (part)

15. By using the substitution $x = u^2$, or otherwise, show that $\int_1^4 \frac{1}{(\sqrt{x})(x+2\sqrt{x}+2)} \, dx = 2 \cot^{-1}(7)$.

Nov 1984/1/22

16. Prove that, for all real x , $\sinh^{-1} x = \ln[x + \sqrt{1 + x^2}]$.

Show that the area of the region lying in the first quadrant and bounded by the y -axis and the curves $y = \frac{1}{\sqrt{2+x^2}}$ and $y = \frac{1}{\sqrt{4-x^2}}$, is $\ln \left[\frac{1}{2} (\sqrt{2} + \sqrt{6}) - \frac{1}{6} \pi \right]$.

Prove also the volume of the solid obtained by rotating this region through one complete revolution about the y -axis is $(4\sqrt{3} - 2\sqrt{2} - 4)\pi$.

MB November 1984/2/20

17. Starting from definitions of $\sinh x$ and $\cosh x$ in terms of e^x , prove that

$$\frac{d}{dx} (\tanh x) = \operatorname{sech}^2 x$$

and that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, ($|x| < 1$).

By means of the substitution $t = \tanh x$, or otherwise, prove that

$$\int \frac{1}{2 \cosh^2 x - \sinh^2 x} \, dx = \frac{1}{\sqrt{2}} \tanh^{-1} \left(\frac{1}{\sqrt{2}} \tanh x \right) + c.$$

MB June 1984/1/20

18. Find

(ii) $\int \sqrt{3-2x^2} dx. \left(\frac{1}{2} x \sqrt{3-2x^2} - \frac{3}{2\sqrt{2}} \sin^{-1} \left(\sqrt{\frac{2}{3}} x \right) + c \right)$

June 1983/1/20

19. (a) Evaluate $\int_0^2 x \tan^{-1} x dx$, giving your answer correct to three decimal places. **(1.768)**

MB November 1983/1/20 (part)

20. (a) Sketch on separate, clearly labelled diagrams the graphs of $y = \tanh x$, ($x \in \mathbb{R}$), and $y = \tanh^{-1} x$, ($x \in \mathbb{R}, |x| < 1$), showing any asymptotes.

(b) Prove that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, $|x| < 1$. Hence or otherwise find

(i) the value of $\frac{d}{dx} (\tanh^{-1} x)$ when $x = \frac{1}{2}$, $\left(\frac{4}{3} \right)$

(ii) $\int_0^{\frac{1}{2}} \tanh^{-1} x dx$. **(0.131)**

MB June 1983/2/20

21. (b) (i) Find $\int \sinh^{-1} x. \left(x \sinh^{-1} x - \sqrt{x^2 + 1} + c \right)$

(ii) Evaluate

$$\int_{\frac{7}{2}}^5 \frac{1}{\sqrt{4x^2 - 16x + 7}} dx,$$

giving your answer to two significant figures. **(0.66)**

December 1982/1/16 (part)

22. (b) By using the substitution $x = a \sin \theta$, or otherwise, show that

$$\int \frac{x^2}{\sqrt{a^2 - x^2}} dx = \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) - \frac{x}{2} \sqrt{a^2 - x^2} + C,$$

where C is an arbitrary constant.

November 1982/2/20 (part)

23. Use the definition $\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$ to prove that $\sinh 2x = 2 \sinh x \cosh x$, and obtain an expression for $\cosh 2x$ in terms of $\sinh x$. **(2 sinh² x + 1)**

By using the substitution $y = \sinh^2 x$, or otherwise, show that

$$\int_0^1 \sqrt{y(1+y)} dy = \frac{1}{4} (3\sqrt{2} - \sinh^{-1} 1).$$

MB November 1982/1/21

24. (a) Find $\int \frac{1}{\sqrt{(x^2+4x-5)}} dx. \left(\cosh^{-1} \left(\frac{x+2}{3} \right) + c \right)$

MB June 1982/1/20 (part)

25. Prove that $(\cosh x + \sinh x)^5 = \cosh 5x + \sinh 5x$ and that $(\cosh x + \sinh x)^5 + (\cosh x - \sinh x)^5 = 2 \cosh 5x$.

Hence, or otherwise, express $\cosh 5x$ as a polynomial of degree five in $\cosh x$.
 $(16 \cosh^5 x - 20 \cosh^3 x + 5 \cosh x)$

From your result, or otherwise,

(a) show that if $u \geq 1$ then $16u^5 - 20u^3 + 5u - 1 \geq 0$,

(b) evaluate $\int_0^{0.4} (5 \cosh^3 x - 4 \cosh^5 x) dx$, **(0.332)**

giving three significant figures in your answer.

November 1981/2/17

26. (a) Evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x dx$, giving your answer correct to 2 significant figures. **(0.13)**

June 1981/2/20 (part)

27. (b) Prove that

$$\int_{\frac{1}{2}\sqrt{2}}^1 \sqrt{(2x^2 - 1)} dx = \frac{1}{2} - \frac{1}{4}\sqrt{2} \cosh^{-1}(\sqrt{2}).$$

MB June 1981/1/20 (part)

28. By using the substitution $y = \sinh x$, or otherwise,

(a) Find $\int \frac{y}{\sqrt{(y^2+1)}} \ln[y + \sqrt{(y^2+1)}] dy$, **$(\sqrt{y^2+1} \ln(y + \sqrt{y^2+1}) - y + c)$**

(b) Show that $\int_0^{\frac{1}{3}} y \ln[y + \sqrt{(y^2+1)}] dy = \frac{41 \ln 3 - 20}{36}$.

MB November 1980/2/21

29. (a) (i) Prove that $\frac{d}{dx} (\sinh^{-1} x) = \frac{1}{\sqrt{(1+x^2)}}$.

(ii) Find $\int \frac{1}{\sqrt{(2x^2-4x+5)}} dx$. **$(\frac{1}{\sqrt{2}} \sinh^{-1}(\frac{x-1}{\sqrt{\frac{3}{2}}}) + c)$**

MB June 1980/2/20 (part)

30. (a) (i) Obtain, as a single hyperbolic function, $\frac{d}{dx} (\coth x)$. **$(-\operatorname{cosech}^2 x)$** [1]

(ii) Prove that $\frac{d}{dx} \left(\frac{1}{\sinh x \cosh x} \right) = -\frac{2}{\sinh^2 x} + \frac{1}{\cosh^2 x \sinh^2 x}$. [2]

(iii) Find $\int \frac{1}{\cosh^2 x \sinh^2 x} dx$. **$(-2 \coth x + \operatorname{cosech} x \operatorname{sech} x + c)$** [4]

(b) Prove that, for $x \neq 0$,

$$\frac{1}{2} + \cosh x + \cosh 2x + \cosh 3x + \cdots + \cosh nx = \frac{\sinh\left(n + \frac{1}{2}\right)x}{2 \sinh \frac{1}{2}x} \quad [7]$$

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31. (b) Find $\int x \operatorname{cosec}^{-1} x dx$. **$(\frac{1}{2} x^2 \operatorname{cosec}^{-1} x + \frac{1}{2} \sqrt{x^2 - 1} + c)$**

November 1979/1/21 (part)

32. Prove that

- (i) $\int \operatorname{sech} x \, dx = 2 \tan^{-1}(e^x) + C,$
 (ii) $\int \operatorname{sech}^3 x \, dx = \frac{1}{2} \tanh x \operatorname{sech} x + \tan^{-1}(e^x) + C,$

where C is, in each case, an arbitrary constant.

November 1979/1/22 (part)

33. (a) Prove that $\int_0^{\ln 2} x \sinh x \, dx = \frac{1}{4}(5 \ln 2 - 3).$

November 1978/1/16 (part)

34. Prove that $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}.$

Hence, or otherwise, prove that $\int_0^{\frac{1}{2}} \tanh^{-1} x = \frac{3}{4} \ln 3 - \ln 2.$

MB November 1978/1/20

- (b) Find $\int \sqrt{x^2 - 4} \, dx. \left(\frac{1}{2} x \sqrt{x^2 - 4} - 2 \cosh^{-1} \left(\frac{1}{2} x \right) + c \right)$

MB June 1978/1/20 (part)

35. Find $\int \sqrt{x^2 - 4} \, dx$ and evaluate $\int_{-1}^1 \sqrt{9x^2 + 1} \, dx$ to three significant figures.
 $\left(\frac{1}{2} x \sqrt{x^2 - 4} - 2 \cosh^{-1} \left(\frac{x}{2} \right) + c; 3.77 \right)$

MB November 1977/2/20 (part)

36. (a) Find the derivative of $\sin^{-1} x + x\sqrt{1-x^2}$, expressing your answer in its simplest form. Hence, or otherwise, evaluate $I = \int_0^{\frac{1}{2}} \sqrt{1-x^2} \, dx. \left(2\sqrt{1-x^2}; \frac{\pi}{2} + \frac{\sqrt{3}}{8} \right)$
 (b) Find $\int x \sin^{-1} x \, dx. \left(\frac{1}{4} (2x^2 - 1) \sin^{-1} x + \frac{1}{4} x \sqrt{1-x^2} + c \right)$

Nov 1976/1/21

37. (a) If $0 < z < a$, find

$$\int_0^z \frac{1}{a^2 - x^2} \, dx.$$

- (i) using partial fractions, $\left(\frac{1}{2} \ln \left(\frac{a+z}{a-z} \right) \right)$
 (ii) using the substitution $x = a \tanh u.$

Deduce an expression for $\tanh^{-1} \left(\frac{z}{a} \right)$ as a natural logarithm.

MB June 1975/2/21 (part)

38. (b) Find

$$\int \frac{1}{\sqrt{x^2 + 2x + 5}} \, dx.$$

$$\left(\ln \left[x + \sqrt{x^2 + 2x + 5} \right] + c \right)$$

MB November 1974/1/22 (part)

39. (c) Find

$$\int_2^3 \frac{1}{\sqrt{(4x^2 - 9)}} dx,$$

giving the answer correct to two significant figures. **(0.81)**

MB November 1973/2/22 (part)

40. Define $\cosh u$, and show that $\cosh u \geq 1$ for all real u .

Write $x = \cosh u$, and deduce a logarithmic form for $\cosh^{-1} x$. ($\ln(x \pm \sqrt{x^2 - 1})$ **OR** $\ln(x + \sqrt{x^2 - 1})$)

Find, in any form,

$$\int \frac{1}{\sqrt{4x^2 - 1}} dx \quad \text{and} \quad \int \sqrt{(x^2 - 1)} dx.$$

$$\left(\frac{1}{2} \cosh^{-1} 2x + c; \frac{1}{2} x \sqrt{x^2 - 1} - \frac{1}{2} \cosh^{-1} x + c \right)$$

MB June 1973/1/20

41. Define $\tanh u$ and show that $-1 < \tanh u < 1$ for all u . Write $x = \tanh u$ and deduce a logarithmic form for $\tanh^{-1} x$. ($\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$)

Find

$$\int \frac{1}{4 - x^2} dx.$$

(a) by using the substitution $x = 2 \tanh u$. $\left(\frac{1}{2} \tanh^{-1} \frac{x}{2} + c = \frac{1}{4} \ln \left(\frac{2+x}{2-x} \right) + C \right)$

(b) by the method of partial fractions. $\left(\frac{1}{4} \ln \left(\frac{2+x}{2-x} \right) + c \right)$

MB November 1972/1/20

(c) Find

$$\int \frac{1}{\sqrt{1 + 4x^2}} dx.$$

$$\left(\frac{1}{2} \sinh^{-1} 2x + c \right)$$

MB November 1972/2/21 (part)